Shlok Mehendale

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Education

BE Computer Science (Hons.), MSc Mathematics (Hons.) 2022 – 2027
BITS Pilani (KK Birla Goa Campus)
Mathematics : Top 5 in the batch; **AWES (National) Scholarship'24 Recipient**

Publications and Preprints

Subspace Crystallization: Geometric and Causal Signatures of Grokking in Modular Arithmetic Networks
Submitted to NeurIPS'26. Poster at LOGML'26, Imperial College London
Authors: ***Shlok Mehendale***, *Paras Chopra (Lossfunk)*
Does representation geometry causally mediate generalization, or do geometric changes merely accompany optimization dynamics that independently drive it?
Selected to present at the **Google Deepmind APAC Research Symposium-2026**

A Radon-Nikodým Perspective on Anomaly Detection: Theory and Implications
Submitted to ACM/IMS Journal of Data Science, 2025. Poster at IndoML'25
Authors: ***Shlok Mehendale***, *Aditya Challa, Rahul Yedida (LexisNexis, USA), Sravan Danda, Santonu Sarkar, Snehanshu Saha* (APPCAIR, BITS Pilani)
This work establishes PAC learnability for anomaly detection and proposes RN-Loss, a loss function weighted by the Radon-Nikodym derivative, that bridges supervised/unsupervised approaches while outperforming SoTA.

Benchmarking Anomaly Detection Algorithms: Deep Learning and Beyond
Submitted to IEEE Transactions on Artificial Intelligence
Authors: *Shanay Mehta** (DevRev), ***Shlok Mehendale***, *Nicole Fernandes** (Grant Thorton, Ireland), *Jyotirmoy Sarkar* (GE Healthcare), *Santonu Sarkar, Snehanshu Saha* (APPCAIR, BITS Pilani)
Cited by **IBM Research.** *= Equal Contribution
This paper benchmarks 30+ anomaly detection algorithms (spanning classical ML, tree-based, and DL methods) across 104 public datasets. It finds that tree-based methods like dBTAI often match or outperform DL, especially on small univariate datasets, while being far faster and requiring no labeled training data.

Matching High-Dimensional Geometric Quantiles for Test-Time Adaptation of Transformers and Convolutional Networks Alike
Submitted to European Conference on Computer Vision (ECCV'26)
Authors: *Sravan Danda, Aditya Challa,* ***Shlok Mehendale***, *Snehanshu Saha* (Learning Theory, APPCAIR, BITS Pilani)
A novel TTA method that explicitly estimates distribution shift between corrupted and clean data by matching high-dimensional geometric quantiles of classifier features, without requiring labeled corrupted data.

Atmospheric Noise Resilient Image Classification: Using a Hybrid Pin-GTSVM
Arxiv Preprint
Authors: ***Shlok Mehendale***, *Jajati Keshari Sahoo, Rajendra Roul* (BITS Pilani)
A hybrid CNN-PinGTSVM model for hazy parking slot classification that eliminates the need for separate dehazing, achieving 98.20% (SoTA) accuracy compared to 88.63% for previous dehazing-based approaches.

Bridging the Gap: A Weighted Attention Bottleneck Transformer for Multimodal Fusion and Reconstruction
University of Toronto ConferenceX'25 proceedings.
Won ProjectX Research competition. Cross-Modal reconstruction, building upon Google Research's work on Multimodal Bottleneck Transformer.

Research Experience

LOGML'26 (Imperial College London) Jun 2026 – Present
Sergio E. Ruiz (Imperial College London, Cambridge, PhD)
Youngest student and only undergrad to be selected for the prestigious London Geometric and Machine Learning summer school. Presented a poster and doing research in singular learning theory.

Visiting Research Scholar - CCVL, Johns Hopkins University Feb 2026 – Present
Prof. Alan L. Yuille (Bloomberg Distinguished Professor of Cognitive Science and Computer Science)
Working on refining the universal subspace hypothesis/the Platonic representation hypothesis and preserving task-specific subspaces in **domain-adaptive continual learning settings.** This is also my undergraduate thesis in the overlap of Learning Mechanics, Mechanistic Interpretability and Cognitive Science.

Lossfunk (Bangalore-based Foundational AI research lab) Dec 2025 – Present
Research Intern (Supervised by Paras Chopra)
Compression along task-relevant directions constrains the hypothesis space and shapes gradient flow, enabling discovery of generalizable solutions. We use these insights to monitor and enable phase transitions in neural networks, such as Grokking. We also study more such rare-phenomenon in neural networks like Neural Collapse.

Undergraduate Student Researcher - APPCAIR (Learning theory) May 2024 - Present
Prof. Snehanshu Saha (Head, APPCAIR)
Developing anomaly detection and imbalanced learning frameworks with applications in **predictive risk modeling** and identifying **outlier financial behaviors**. Engineered a robust statistical loss function grounded in measure theory, applicable to **stochastic loss forecasting** and **risk management** frameworks.

Network Security for Smart and Secure Buildings in Sweden Jul 2024 - Nov 2024
Dr. Christer Ahlund, Dr. Karan Mitra, Dr. Saguna (LTU Sweden), Dr. Neena Goveas (BITS Goa)
Industry funded project, deployed in **Sweden (Ace-cybersafe)**. Worked with BrickSchema-modeled IoT sensor data to detect and classify faults. Integrated semantic modeling and real-time data streams for fault monitoring.

Project Kratos (A Mars Rover Team) Aug 2023 - Jan 2025
Mars Rover Team, BITS Goa
Deep Learning based Hand Gesture Recognition Controlled Rover (Individual project) using a Gated LSTM. Achieved probabilistic mapping, worked with point cloud data and elevation maps using Zed 2i stereo camera. Led and set up a collaboration and a thriving community in a competitive environment with AGH Space System (Poland).

Confluence of Epidemiological Models and Deep Learning
BITS Pilani (KK Birla Goa Campus) and Yale University
Presented to Dr. Abhishek Pandey, Associate Director (CIDMA), Yale. Modified **Epi-DNNs** for **Indian Covid-19 data** and a novel SEIRD model. Won the course's best presentation award.

Minor Projects
- **CompreRAG** : Developed an LLM-based RAG agent for complex document ingestion, demonstrating potential for **automating regulatory report analysis** and data extraction. BETA version: **AI Teaching Assistant @Aden(YC'20)**
- **Shearlet-based X-ray tomography**: **Best presentation** (Functional Analysis). Implemented the paper "Shearlet-based regularization in statistical inverse learning with an application to x-ray tomography" TA Bubba, Luca Ratti; and compared with modern reconstruction techniques using Diffusion models, NeRFs
- **IMAGraph: Image Manipulation through Adjacency Graphs** : Dynamically generated and altered adjacency matrices of CIFAR 10 images, observing changes in Image processing and generation
- **A Sorting Assignment** : A modular Python application that demonstrates separation of concerns across sorting, I/O, networking, and orchestration layers. A simple project blown completely out of proportions

Awards and Recognition

UofT AI Research Competition Winner (ProjectX)
Team Leader, University Of Toronto, Canada
Represented BITS at an international level and led a team of Cs undergrads. Featured in **NavHind Times** (national newspaper). Participating teams included ETH Zurich, Cornell, University of Edinburgh, etc.

NXP AIM Hackathon (National and Regional Winners)
NXP Semiconductors, Time of Sports
Developed quantised models for sign recognition and obstacle avoidance for autonomous vehicles.

Amazon ML Challenge'25 (Top 1%)
Developed a **Gated Multimodal Fusion Network** focusing on text representation learning. Integrated BGELarge text embeddings with structured and image features via gated attention, enabling adaptive weighting of modalities.

Additional Information
- Ranked 3rd in the International Rover Challenge'24, featured in **Times of India**
- Selected for research presentation to Olaf Scholz, **former German Chancellor**
- Selected as an **IKDD Uplink intern** under Dr. Vineeth Balasubramanian (**Microsoft Research**, IIT Hyderabad)
- One of the few undergraduates selected for **Brain, Computation and Learning Workshop'25 (IISc Bangalore); CVIT Summer School'24,'25; 3D-Vision Summer School'25 (IIIT Hyderabad)**

Teaching Experience

Teaching Assistant: Mathematics-2 (Linear Algebra, Complex Numbers): Semester-2 '23-24, Semester-2 '24-25
Probability and Statistics: Semester-1 '24-25
WILP (Coursera): Command Line Scripting: Semester-2 '24-25; Object-oriented Programming: Semester-1 '25-26; Data Visualization: Semester-2 '25-26

STEM4ALL (California): Designed a course on Deep Learning for working professionals in SF, California

Founding Experience

PathFinders Jan 2023 - Dec 2023
Founder (Executive Head)
A not-for-profit organization for mentoring and guidance of engineering aspirants. Led a small team of undergraduates across colleges in India, mentoring 28 students, leading to the finest institutes.

Henneth Nov 2024 - Feb 2025 | Stanford, USA
Co-Founder (Technical)
Co-founded AI startup developing autonomous testing agents to replace beta testing phase in software dev cycle.